



Required Capstone Project: Module 3

Due Jun 3 by 11:59pm**Points** None**Available** after May 20 at 12am

☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186764)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186766)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186767)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186768)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186769)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186770)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186771)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186772)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186773)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186774)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186775)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186776)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186777)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186778)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186779)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186780)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186781)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186783)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186785)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186786)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186787)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186788)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186789)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186790)

☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186791)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186792)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186793)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186794)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186795)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186796)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186797)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186798)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186799)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186800)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186802)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186803)
☒		(https://classroom.emeritus.org/courses/16839/modules/items/3186805)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186903)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3186808)
☐		(https://classroom.emeritus.org/courses/16839/modules/items/3201439)

Learning Outcomes Addressed

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 1 – Problem Framing and Generative AI Foundations

Milestone 3: Prompt & Output Evaluation Toolkit

Estimated Effort: 1.5 Hour

Word Limit: 1000 words

Objective

In **Module 1**, you selected a business workflow and created an **AI Opportunity Brief** identifying a potential use case for Generative AI.

In **Module 2**, you analysed that workflow using a **Workflow Baseline & Risk Canvas**, documenting how the process currently operates and identifying tasks where Generative AI could potentially assist.

In this module, you will move from **workflow analysis to practical AI interaction** by designing structured prompts that simulate how Generative AI could assist with specific tasks in that workflow.

You will create a **Prompt & Output Evaluation Toolkit**, which demonstrates how prompts can guide Generative AI systems to produce useful, structured, and reviewable outputs.

The toolkit will include **five prompts representing different types of Generative AI tasks**, along with clear input structures, output formats, validation criteria, and human review rules.

These prompts should ideally correspond to **tasks or inefficiencies identified in your workflow analysis in Module 2**.

For example, if your workflow involved reviewing documents, analysing messages, preparing reports, or validating information, the prompts you design should simulate how AI could assist with those activities.

The prompts created in this module will later serve as **building blocks for AI-enabled workflow redesign in later capstone modules**.

Note: This is a required assignment and counts towards programme completion.

Task Overview

Using the **workflow you selected in Module 1 and analysed in Module 2**, design **five structured prompts** that demonstrate how Generative AI could assist with tasks inside that workflow.

Each prompt should represent a **distinct Generative AI capability**.

Your prompts should demonstrate the following types of AI tasks:

1. **Summarisation**
2. **Classification**
3. **Decision Support**
4. **Structured Reporting**
5. **Risk-Mitigation Prompt**

The objective is not only to write prompts, but to design them in a way that ensures:

- clear input structure
- consistent output formats
- reliable evaluation of AI responses
- appropriate human oversight

If Your Workflow Does Not Include All Prompt Types

Not every workflow naturally includes all five task types listed above.

For example, some workflows may involve:

- operational coordination
- structured software processes
- transactional data entry

In such cases, your goal is **not to replicate the current workflow exactly**, but to design prompts that **demonstrate how Generative AI could assist or enhance the workflow**.

The prompt categories represent **AI capabilities**, not necessarily existing workflow steps.

For example:

Prompt Type	How It Could Support a Workflow
Summarisation	Summarising emails, notes, or documents
Classification	Categorising requests, tickets, or document types
Decision Support	Highlighting recommendations or priority actions
Structured Reporting	Converting unstructured updates into structured summaries
Risk-Mitigation Prompt	Reviewing outputs and identifying potential errors

If your workflow does not currently contain one of these tasks, you may design a prompt that **supports the workflow indirectly**, such as summarising communications, generating structured updates, or validating outputs.

The goal is to demonstrate **how prompting techniques enable Generative AI to assist real business processes**.

Step 1 — Identify Workflow Tasks Suitable for AI Assistance

Review the workflow analysis you completed in **Module 2**.

Look for steps where employees must:

- read large amounts of text
- extract information from documents
- classify incoming requests or documents
- analyse information to make decisions
- prepare summaries or reports

These tasks often represent **strong opportunities for Generative AI assistance**.

The prompts you design in this module should ideally correspond to **these workflow activities**.

Example mapping:

Workflow Activity	Potential AI Prompt
Reviewing long documents	Summarisation prompt
Sorting incoming requests	Classification prompt
Evaluating proposals	Decision support prompt
Preparing updates or reports	Structured reporting prompt
Verifying AI outputs	Risk-mitigation prompt

Step 2 — Design Prompts for Five AI Task Types

Create **five prompts**, each representing a different Generative AI capability.

These prompts should simulate how employees might interact with AI systems to complete workflow tasks more efficiently.

1. Summarisation Prompt

Design a prompt that asks the AI system to summarise large or complex information.

Typical workflow uses include:

- summarising documents
- condensing meeting transcripts
- extracting insights from reports

Example task:

Summarise a vendor contract into key clauses and obligations.

2. Classification Prompt

Design a prompt that categorises or labels incoming information.

Common business uses include:

- classifying customer emails
- tagging support tickets
- identifying document types

Example task:

Classify incoming customer queries into categories such as billing, technical support, or product inquiry.

3. Decision Support Prompt

Design a prompt where the AI analyses information and provides structured recommendations.

Decision-support prompts help employees evaluate options more efficiently.

Example task:

Analyse vendor proposals and highlight the most cost-effective option based on predefined criteria.

4. Structured Reporting Prompt

Design a prompt that converts unstructured information into a **structured output format**.

Common business uses include:

- generating structured reports
- extracting metrics from documents
- producing formatted summaries for dashboards

Example task:

Extract key project updates from status reports and format them into a structured weekly summary.

5. Risk-Mitigation Prompt

Design a prompt that explicitly reduces AI risk by introducing **verification or caution rules**.

These prompts help reduce issues such as hallucinations or unsupported claims.

Example task:

Review a draft financial summary and highlight statements that cannot be verified from the provided data.

Step 3 — Define Input Structure

For each prompt, specify **what information will be provided to the AI system**.

Inputs may include:

- documents
- text messages
- emails
- reports
- data tables

Clearly defining inputs improves **prompt reliability and output consistency**.

Example input structure:

Customer email text

Customer account information

Previous support ticket history

Step 4 — Define the Output Format

Specify how the AI should present its response.

Structured outputs make AI responses easier to review and integrate into workflows.

Examples include:

- bullet-point summaries
- labelled categories
- tables
- decision recommendations with reasoning

Example output structure:

Summary

Key Issues Identified

Recommended Next Action

Step 5 — Define Validation Criteria

Each prompt should include **criteria for evaluating whether the AI output is acceptable**.

Validation criteria ensure outputs remain **reliable and aligned with business requirements**.

Examples include:

- outputs must reference only the provided information
- summaries must remain within a defined length
- classifications must use predefined categories

Example validation rule:

The summary must only include information present in the source document.

Step 6 — Define a Human Review Rule

Generative AI outputs should not be accepted without verification.

For each prompt, specify **how a human will review the AI response**.

Human review may involve:

- verifying factual accuracy
- confirming classifications
- approving recommendations before action is taken

Example:

Finance team must verify extracted invoice details before they are entered into the accounting system.

Required Prompt Template

For each of the **five prompts**, include the following elements.

Prompt Purpose

Describe the workflow task the prompt supports.

Prompt Text

Write the full prompt that would be provided to the AI system.

Input Structure

Describe the information provided to the model.

Expected Output Format

Explain how the AI response should be structured.

Validation Criteria

Define how the output will be evaluated.

Human Review Rule

Explain how employees will verify the AI output.

Practical Tips

Connect Prompts to Your Workflow

- ✓ Design prompts that relate to the **workflow analysed in Modules 1 and 2**
 - ✓ Focus on real operational tasks where AI could assist
-

Use Clear Instructions

- ✓ Specify the task clearly
- ✓ define input data
- ✓ define expected output structure

Clear prompts produce **more consistent AI responses**.

Design Prompts for Operational Use

Strong prompts should be:

- clear
- repeatable
- easy to evaluate

Think about how the prompt would function **inside a real organisational workflow**.

Example Prompt (Simplified)

Prompt Purpose

Summarise a customer complaint email.

Prompt Text

Summarise the following customer complaint in three bullet points. Identify the main issue, the customer request, and any urgency indicators.

Input Structure

Customer email text.

Expected Output Format

Main Issue

Customer Request

Urgency Level

Validation Criteria

The summary must only use information present in the email.

Human Review Rule

Customer support agent must verify the summary before sending a response.

Submission

Before submitting your canvas, confirm that you have included:

- ✓ Five structured prompts
- ✓ One prompt for each required task category
- ✓ Input structures for each prompt
- ✓ Expected output formats
- ✓ Validation criteria for outputs
- ✓ Human review rules
- ✓ Prompts connected to tasks identified in your workflow

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 15

Criteria	Full Points	0 Points
Problem Understanding &	The submission explains the business problem, demonstrates understanding	Problem complexity is not explained. No justification

Criteria	Full Points	0 Points
Need for Advanced Prompting (3 points)	of its complexity, and justifies why basic prompting is insufficient, highlighting the need for structured reasoning. (3 points)	for using advanced prompting techniques. (0 points)
Prompt Design & Advanced Techniques Application (4 points)	The submission includes prompts (CoT, Few-shot CoT, decomposition) and applies at least two advanced techniques (e.g. ToT, ReAct, self-consistency). (4 points)	Prompts are missing, incomplete, or do not demonstrate use of advanced techniques. (0 points)
Workflow Design, Guardrails & Output Control (4 points)	The submission presents a multi-step prompt workflow (pipeline), including logical sequencing. Guardrails (constraints, hallucination control) and structured output formats (e.g. JSON/table) are defined. (4 points)	Workflow is missing. No guardrails, constraints, or output structure defined. (0 points)
Evaluation, Accuracy & Trade-offs (4 points)	The submission compares techniques based on accuracy, consistency, and reliability. Identifies the best-performing approach, key improvements, and clearly explains trade-offs (e.g. cost, latency, complexity). (4 points)	No evaluation or comparison provided. Performance, improvements, and trade-offs are not discussed. (0 points)

Why this Step Matters

Generative AI systems are most effective when prompts are **structured, repeatable, and aligned with business workflows**.

By designing prompts connected to your workflow analysis, you learn how to:

- translate operational tasks into AI interactions
- guide AI systems toward consistent outputs
- introduce validation and governance controls
- prepare AI prompts for real organisational workflows

In later modules, these prompts will help you **design AI-enabled workflow improvements and automation systems.**